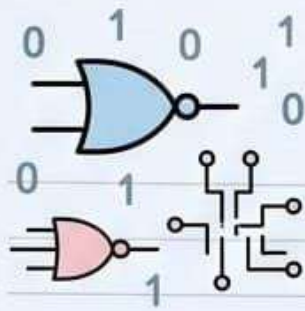
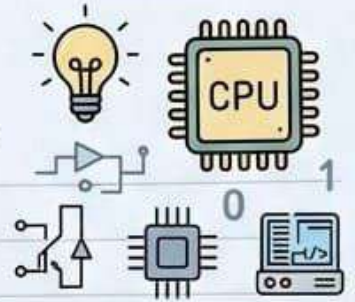




NOT Gate

NOT ද්වාරය

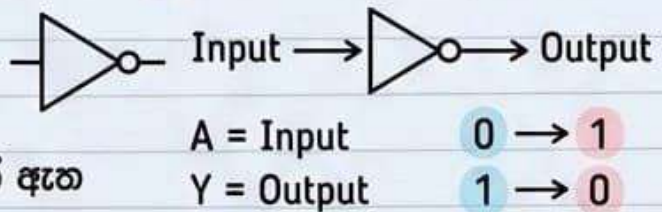


NOT Gate යනු input එකේ අගය ප්‍රතිවිරුද්ධ output එකක් ලබාදෙන Digital Logic Gate එකකි.

01. NOT Gate යනු කුමක්ද?

- Digital Logic Gate එකකි
- Inverter ලෙසද හැඳින්වේ
- Input එක reverse කරයි
- Single input එකක් පමණක් ඇත

02. NOT Gate Symbol



03. Truth Table

Input (A)	Output (Y)
0	1
1	0

Input 0 නම් output 1 වේ.
Input 1 නම් output 0 වේ.

04. Boolean Expression

$$Y = \bar{A} \text{ or } Y = \text{NOT } A$$

Boolean Algebra වල NOT operation එක inverse කිරීම සඳහා භාවිතා වේ.

05. Working Principle

Input Signal $\rightarrow$ NOT Gate $\rightarrow$ Opposite Output

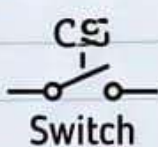
Switch OFF $\rightarrow$ Bulb ON

Switch ON $\rightarrow$ Bulb OFF

06. Uses of NOT Gate

- ✓ Signal inversion
- ✓ Memory systems
- ✓ Digital circuits
- ✓ Control systems
- ✓ Computers
- ✓ Logic operations

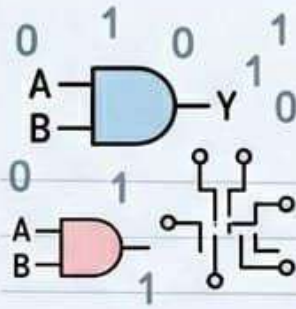
07. Real-Life Example



- Automatic control systems
- Alarm systems
- Sensor circuits
- Computer processors

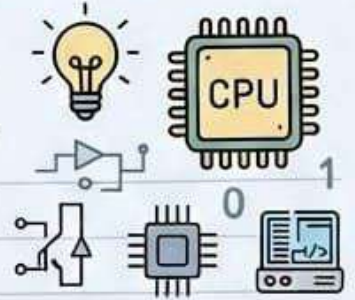
- ✓ NOT Gate = Inverter
- ✓ Single input logic gate
- ✓ Output always opposite
- ✓ Boolean expression $\rightarrow Y = \bar{A}$

NOT Gate එක input එකේ ප්‍රතිවිරුද්ධ output ලබාදෙන මූලික Digital Logic Gate එකකි.



AND Gate

AND ද්වාරය

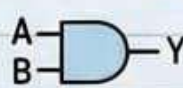


AND Gate යනු inputs සියල්ලම 1 වූ විට පමණක් output එක 1 ලබාදෙන Digital Logic Gate එකකි.

01. AND Gate යනු කුමක්ද?

- Digital Logic Gate එකකි
- Inputs දෙකක් හෝ වැඩි
- සියලු inputs 1 නම් පමණක්
- Basic logic operation එකකි

02. AND Gate Symbol



A, B = Inputs

Y = Output

1 AND 1 = 1

1 AND 0 = 0

03. Truth Table

A	B	Y
0	0	0
0	1	0
1	0	0
1	1	1

Inputs දෙකම 1 වූ විට පමණක් output 1 වේ.

04. Boolean Expression

$$Y = A \cdot B \quad Y = A \text{ AND } B$$

Boolean Algebra වල multiplication operation එක AND Gate එක නිරූපණය කරයි.

05. Working Principle

Switch A → AND Gate → Bulb
Switch B

OFF + OFF → OFF

ON + OFF → OFF

ON + ON → ON

06. Uses of AND Gate

✓ Digital circuits

✓ Security systems

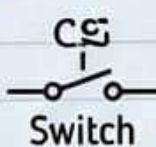
✓ Computers

✓ Control systems

✓ Arithmetic circuits

✓ Decision making circuits

07. Real-Life Example



- Two-switch light systems
- Alarm systems
- Industrial automation
- Computer processors

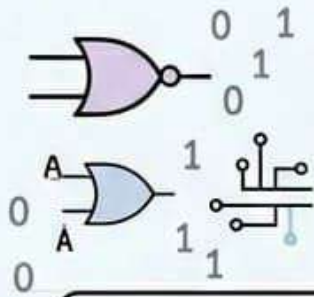
✓ Output 1 only when all inputs are 1

✓ Basic Digital Logic Gate

✓ Boolean expression → $Y = A \cdot B$

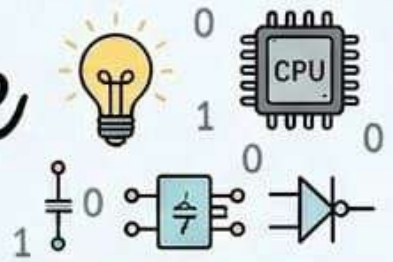
✓ Used in digital electronics

AND Gate එක inputs සියල්ලම 1 වූ විට පමණක් output 1 ලබාදෙන මූලික Digital Logic Gate එකකි.



NOR Gate

NOR ද්වාරය



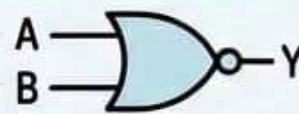
NOR Gate යනු OR Gate එකේ output එක inverse කරන Digital Logic Gate එකකි.

01. NOR Gate යනු කුමක්ද?

- Digital Logic Gate එකකි
- OR + NOT combination එකකි
- Inputs සියල්ලම 0 වූ විට පමණක් output 1 වේ
- Universal Gate එකක් ලෙස හැඳින්වේ



02. NOR Gate Symbol



A, B = Inputs
Y = Output

0 NOR 0 = 1
1 NOR 0 = 0

03. Truth Table

A	B	Y
0	0	1
0	1	0
1	0	0
1	1	0

Inputs සියල්ලම 0 වූ විට පමණක් output 1 වේ.

04. Boolean Expression

$$Y = \overline{A + B}$$

$$Y = \text{NOT } (A \text{ OR } B)$$

Boolean Algebra වල OR operation එක inverse කිරීම NOR Gate එකෙන් සිදු වේ.

05. Working Principle



OFF + OFF → ON
ON + OFF → OFF
ON + ON → OFF

06. Uses of NOR Gate

- ✓ Computers
- ✓ Digital circuits
- ✓ Memory systems
- ✓ Logic circuit design
- ✓ Arithmetic circuits
- ✓ Universal gate circuits

07. Universal Gate

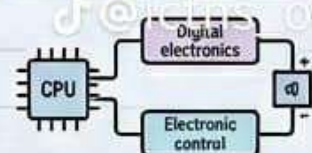
- NOR Gate පමණක් භාවිත කර වෙනත් gates නිර්මාණය කළ හැක
- NOT, OR, AND gates build කළ හැක



KEY POINT STICKY NOTE

- ✓ NOR = OR + NOT
- ✓ Universal Logic Gate
- ✓ Output 1 only when all inputs are 0
- ✓ Boolean expression
→ $Y = (A + \overline{B})$

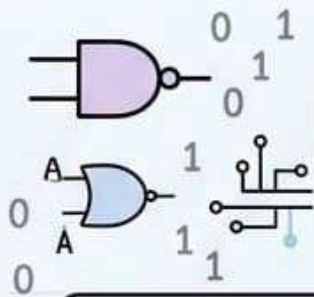
08. Real-Life Example



- Computer processors
- Automation systems
- Digital electronics
- Embedded systems

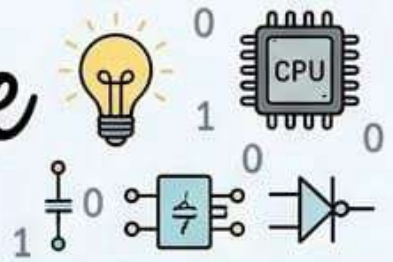
NOR Gate එක OR Gate output එක inverse කරන Universal Digital Logic Gate එකකි.

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NAND Gate

NAND ද්වාරය



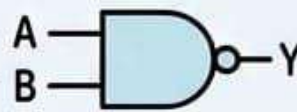
NAND Gate යනු AND Gate එකේ output එක inverse කරන Digital Logic Gate එකකි.

01. NAND Gate යනු කුමක්ද?

- Digital Logic Gate එකකි
- AND + NOT combination එකකි
- Inputs සියල්ලම 1 වූ විට පමණක් output 0 වේ
- Universal Gate එකක් ලෙස හැඳින්වේ



02. NAND Gate Symbol



A, B = Inputs
Y = Output
1 NAND 1 = 0
1 NAND 0 = 1

03. Truth Table

A	B	Y
0	0	1
0	1	1
1	0	1
1	1	0

Inputs දෙකම 1 වූ විට පමණක් output 0 වේ.

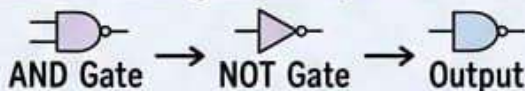
04. Boolean Expression

$$Y = \overline{A \cdot B}$$

$$Y = \text{NOT } (A \text{ AND } B)$$

Boolean Algebra වල AND operation එක inverse කිරීම NAND Gate එකෙන් සිදු වේ.

05. Working Principle



ON + ON → OFF
ON + OFF → ON
OFF + OFF → ON

06. Uses of NAND Gate

- ✓ Computers
- ✓ Digital circuits
- ✓ Memory systems
- ✓ Logic circuit design
- ✓ Arithmetic circuits
- ✓ Universal gate circuits

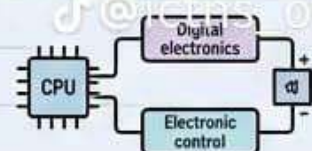
07. Universal Gate

- NAND Gate පමණක් භාවිතා කර වෙනත් gates නිර්මාණය කළ හැක
- NOT, AND, OR gates build කළ හැක



- ✓ NAND = AND + NOT
- ✓ Universal Logic Gate
- ✓ Output 0 only when all inputs are 1
- ✓ Boolean expression
→ $Y = (A \cdot \overline{B})$

08. Real-Life Example



- Computer processors
- Automation systems
- Digital electronics
- Embedded systems

NAND Gate එක AND Gate output එක inverse කරන Universal Digital Logic Gate එකකි.

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